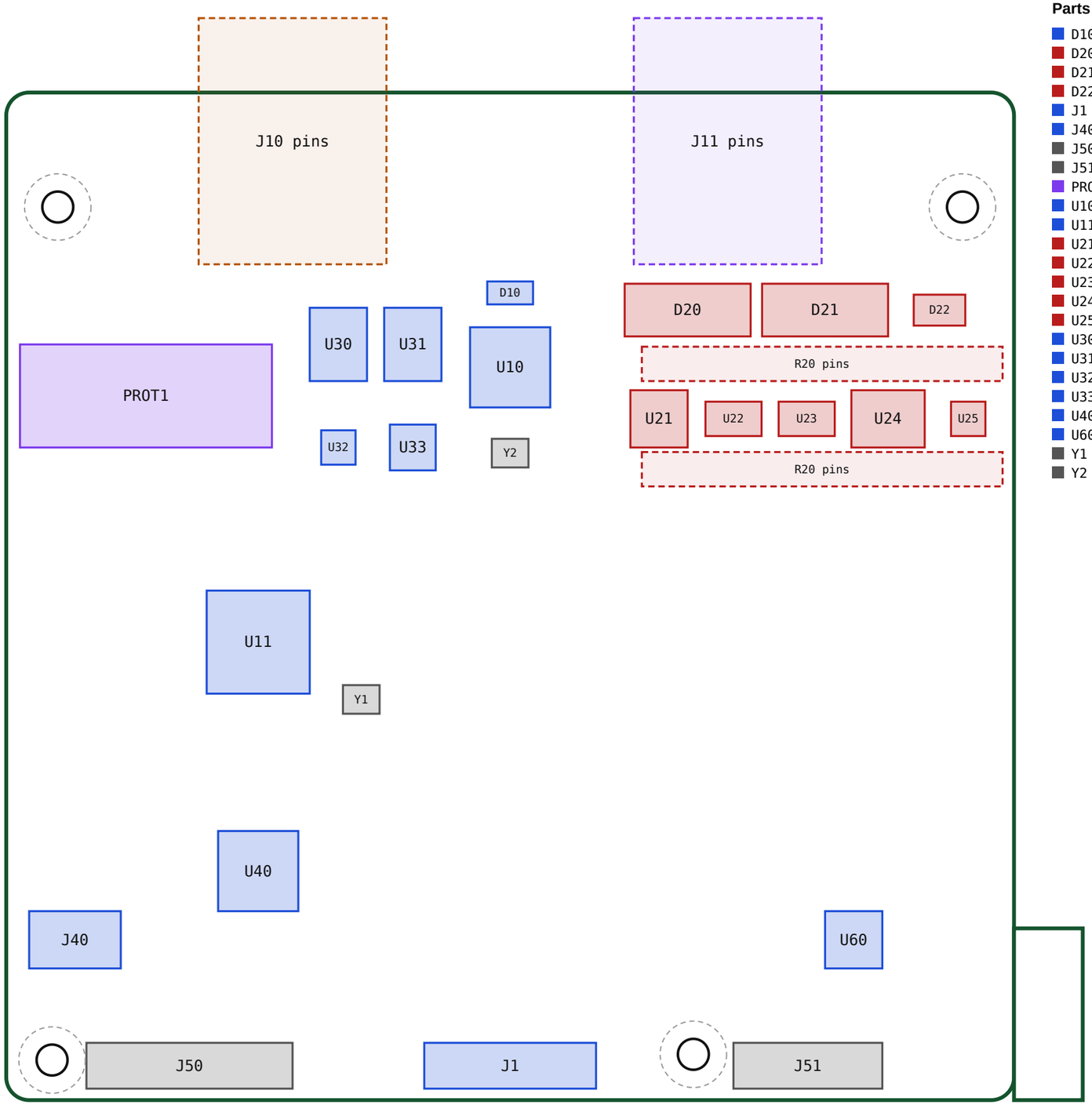
Main board: 88 x 88 mm + card tab, 6 layers - FRONT (display side)

Floorplan only: not routed, approximate envelopes, small passives not placed.



Parts on this side (height mm)

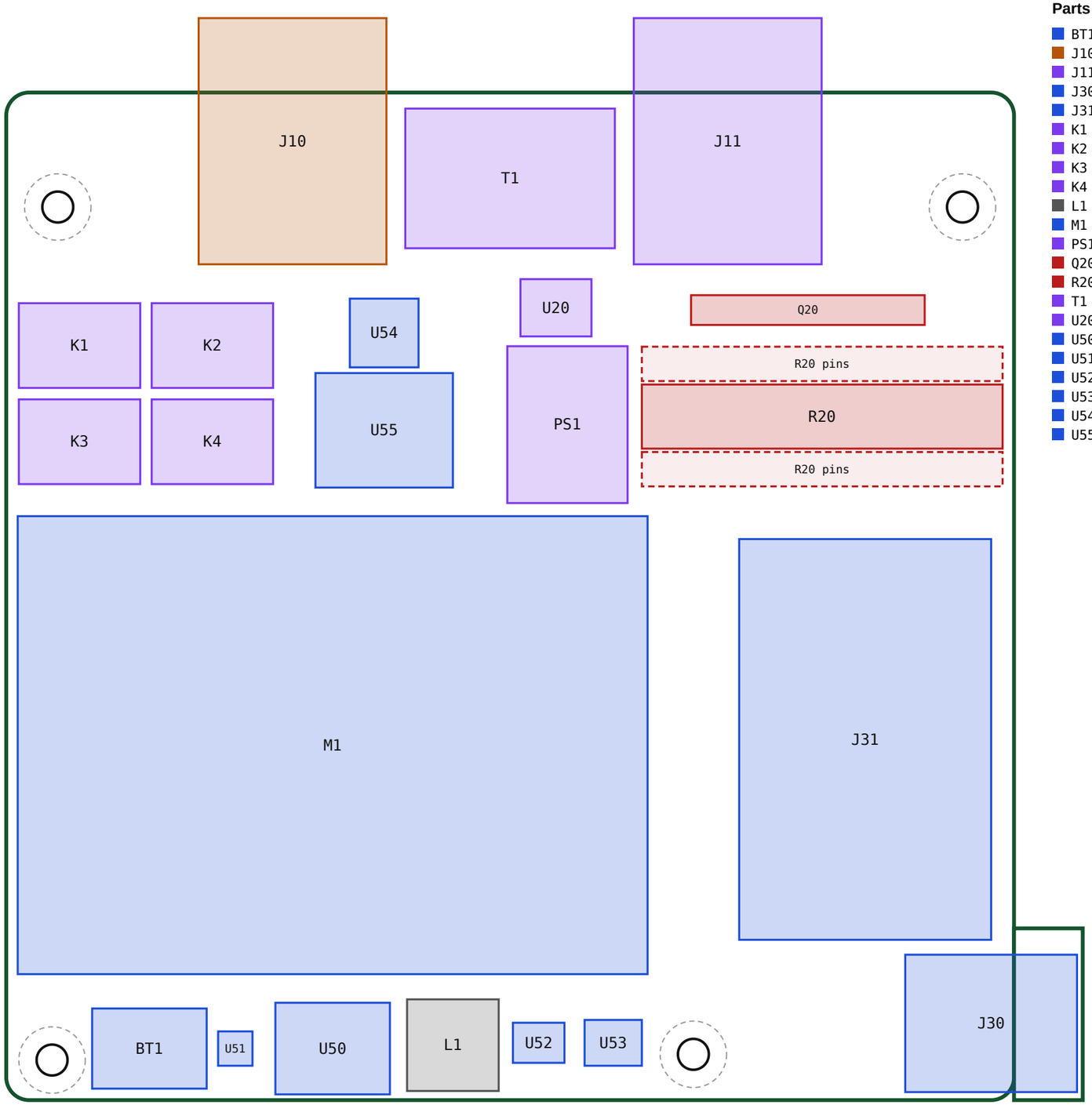
- D10 0.6 PHY-side TVS array x2
- D20 1.1 Schottky bridge, Alt A
- D21 1.1 Schottky bridge, Alt B
- D22 1.1 TVS 58 V
- J1 1.0 DSI FFC to 5in touch display
- J40 0.0 debug: UART + SWD + boot header
- J50 1.0 (not a netlist part)
- J51 1.0 (not a netlist part)
- PROT1 1.2 8-line protection network
- U10 0.9 KSZ9131RNX gigabit PHY
- U11 0.9 LAN7801 USB3-to-RGMII MAC
- U21 0.9 TPS2372-4 PD interface
- U22 1.1 INA228 V/I/P monitor
- U23 1.1 ADS1115 centre-tap voltage ADC
- U24 1.2 TCA9534 load-step expander
- U25 1.2 over-temp comparator + NTC
- U30 1.2 ADG5408 8:1 mux (drive)
- U31 1.2 ADG5408 8:1 mux (return/sense)
- U32 1.1 precision op-amp, source + sense
- U33 0.9 crosstalk tone driver/detector
- U40 0.6 STM32G0B1 coprocessor
- U60 0.9 USB2244 microSD reader (DNP option)
- Y1 0.8 (not a netlist part)
- Y2 0.8 (not a netlist part)

Blue system side, amber cable conductors, red network-powered, purple isolation barrier. Dashed = through-hole pins. Scale 9 px/mm.

10 mm

Main board: 88 x 88 mm + card tab, 6 layers - BACK (seen through the board)

Floorplan only: not routed, approximate envelopes, small passives not placed.



Parts on this side (height mm)

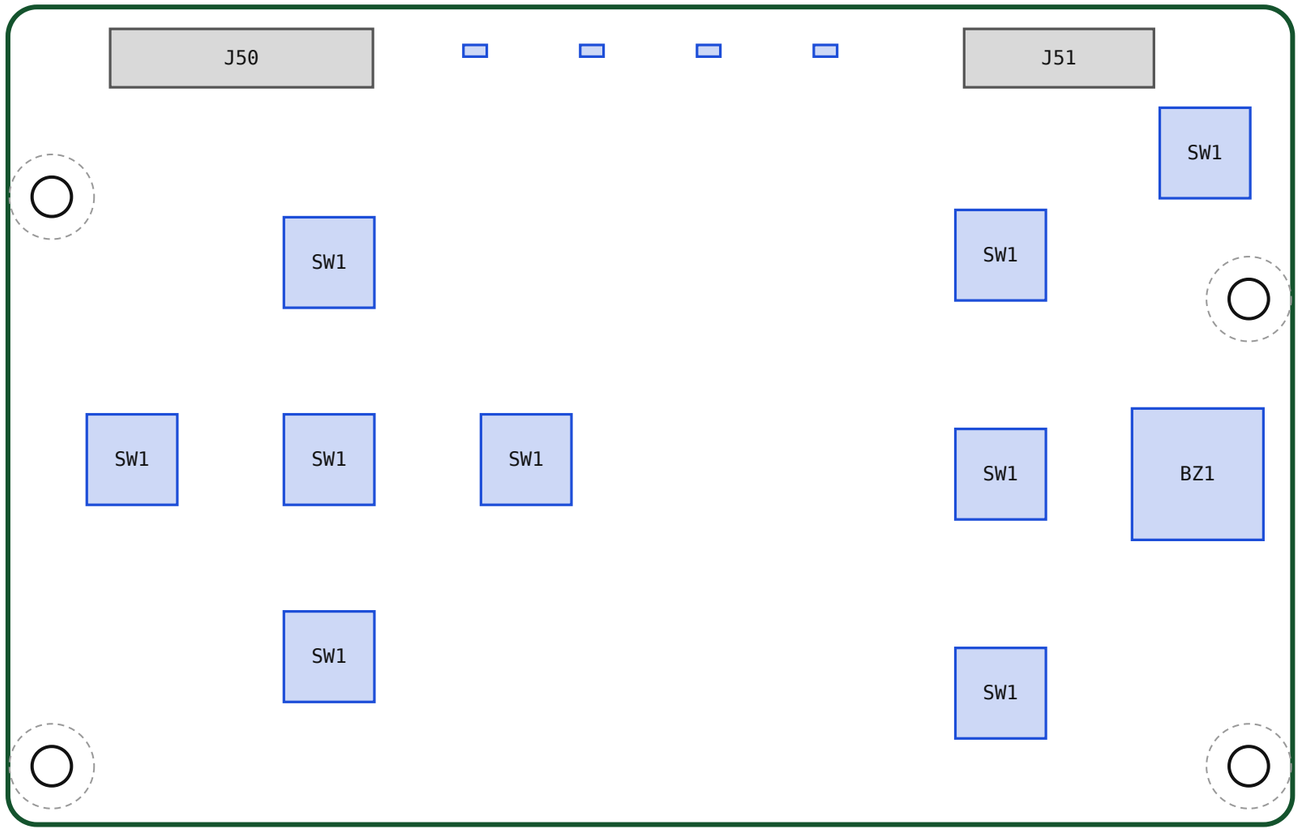
- BT1 6.0 1S3P 21700 pack with PCM + NTC
- J10 13.5 RJ45 shielded, no magnetics
- J11 13.5 RJ45 with integrated 1G magnetics
- J30 1.9 microSD socket (DNP option)
- J31 4.2 M.2 E-key 2230 (DNP option)
- K1 5.4 DPDT signal relay, pair 1-2
- K2 5.4 DPDT signal relay, pair 3-6
- K3 5.4 DPDT signal relay, pair 4-5
- K4 5.4 DPDT signal relay, pair 7-8
- L1 4.0 (not a netlist part)
- M1 6.0 Raspberry Pi CM5 wireless
- PS1 3.2 isolated 5 V DC/DC, 1 W
- Q20 1.1 load-step MOSFETs x6
- R20 19.0 load resistor bank x6 on spreader
- T1 6.4 1000BASE-T PoE magnetics, 4 CT
- U20 1.75 ISO1640 I2C isolator
- U50 2.0 BQ25895 1S charger
- U51 0.8 MAX17055 fuel gauge
- U52 0.9 TPS61088 5 V boost
- U53 1.2 TPS62840 3V3 always-on buck
- U54 2.0 3V3 system LDO/buck for PHY + AFE
- U55 3.0 +/-15 V charge pump for mux rails

Blue system side, amber cable conductors, red network-powered, purple isolation barrier. Dashed = through-hole pins. Scale 9 px/mm.

10 mm

Keypad and USB-C board: 88 x 56 mm, 2 layers - FRONT (display side)

Floorplan only: not routed, approximate envelopes, small passives not placed.



Parts on this side (height mm)

- BZ1 3.0 piezo buzzer
- J50 1.0 (not a netlist part)
- J51 1.0 (not a netlist part)
- LED1 0.6 status LEDs x4
- SW1 9.5 8-key keypad + power key

Blue system side, amber cable conductors, red network-powered, purple isolation barrier. Dashed = through-hole pins. Scale 9 px/mm.

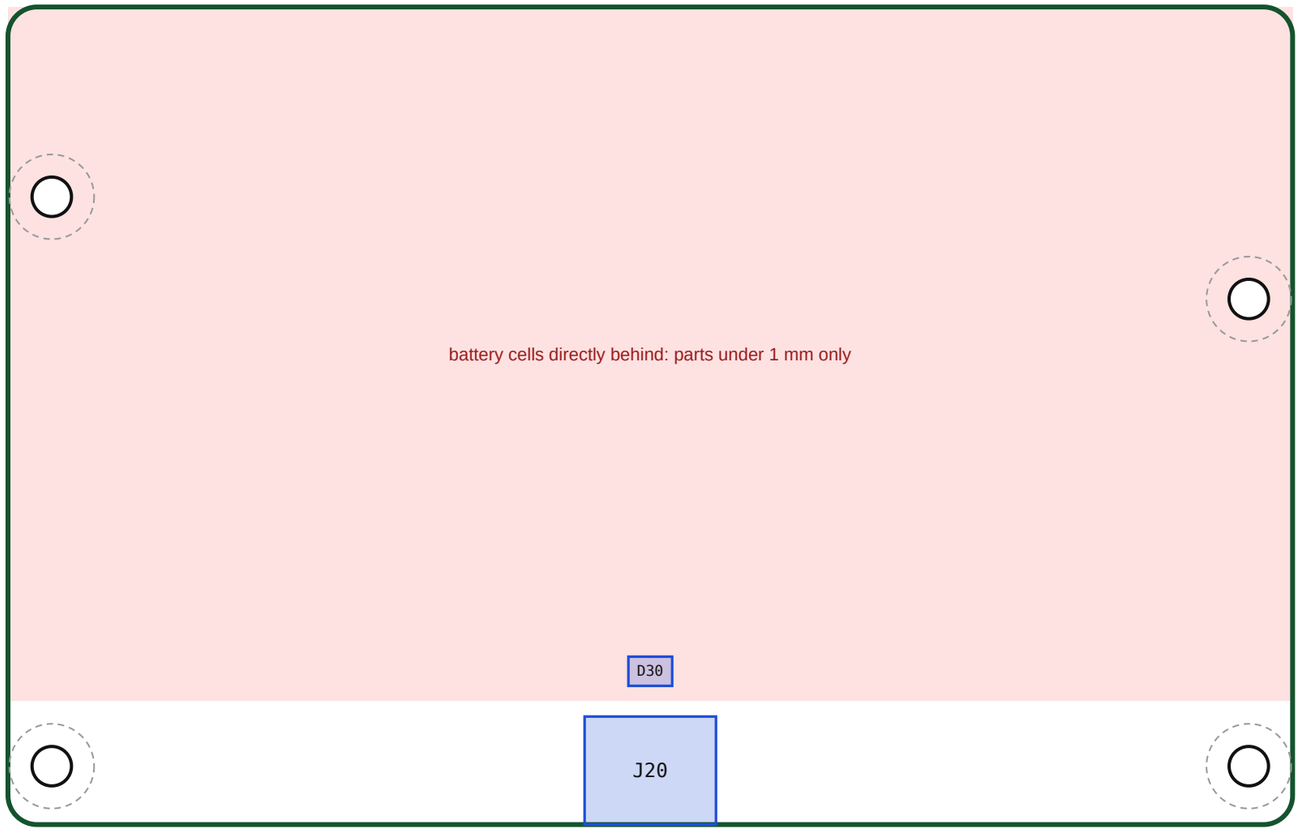
10 mm

Keypad and USB-C board: 88 x 56 mm, 2 layers - BACK (seen through the board)

Floorplan only: not routed, approximate envelopes, small passives not placed.

Parts on this side (height mm)

- D30 0.6 USB ESD array
- J20 3.3 USB-C receptacle



Blue system side, amber cable conductors, red network-powered, purple isolation barrier. Dashed = through-hole pins. Scale 9 px/mm.

10 mm